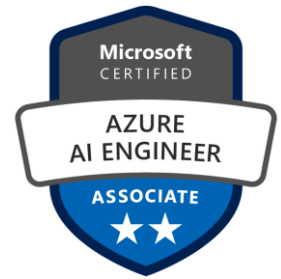


Niharika Kollikonda
Gen-AI/AI Engineer

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Professional Summary:

- AI Engineer with 9+ years of experience in Python, Machine Learning, and Data Engineering. Expert in designing scalable ML pipelines and integrating advanced technologies such as Google Cloud LLMs, Google Vector Search, and Hybrid Search methods. Skilled in AWS SageMaker, Docker, Django, and Flask, with a proven record of deploying production-grade ML solutions and optimizing MLOps workflows for large-scale distributed systems.
- Experienced in developing and deploying advanced search pipelines leveraging Google Cloud LLMs, Google Vector Search, and Hybrid Search Techniques.
- Hands-on expertise in Python, PyTorch, TensorFlow, and Keras, with strong proficiency in NLP frameworks such as SpaCy, NLTK, and Hugging Face Transformers. Deep implementation experience with large language models including GPT, LLaMA, Claude, BERT, RoBERTa, and LangChain. Skilled in Vector Databases, LoRA, RAG, PEFT, and Knowledge Graphs.
- Proficient in the AWS ecosystem (SageMaker, EC2, Lambda, EMR, S3) with extensive experience in Jupyter Notebooks, agentic programming, and AI frameworks such as GraphRAG, Chain of Thought (CoT), and Tree of Thought (ToT). Expert in managing structured and unstructured data, deploying models, and delivering high-impact AI/ML solutions using advanced data science tools.
- Strong background in NLP and Generative AI, including GPT, LLaMA, Claude, BERT, RoBERTa, and custom model development using Hugging Face Transformers, SpaCy, and NLTK. Skilled in RAG, LoRA, PEFT, and Knowledge Graphs for intelligent retrieval and reasoning.
- Experienced in Explainable AI (XAI) and production-grade LLM customization for agentic and reasoning-based AI systems (GraphRAG, Chain of Thought, Tree of Thought).
- Solid data engineering foundation with expertise in SQL, Spark, Linux, Git/GitHub, and ETL tools like Informatica PowerCenter. Skilled in Data Migration, Profiling, Cleansing, Transformation, and Integration for large-scale data ecosystems.
- Strong data visualization and analytics expertise, creating interactive dashboards and reports using Tableau, Power BI, QuickSight, Kibana, Python Matplotlib, and R Shiny.
- Comprehensive knowledge of Data Modeling and Warehousing, applying Star, Snowflake, and Extended Star schemas. Skilled in building ER diagrams, schema designs, and managing structured/unstructured datasets.
- Deep understanding of statistical and mathematical modeling, including hypothesis testing (t-tests, ANOVA, z-scores, p-values), model evaluation (confusion matrix, ROC, AUC, precision, recall), and predictive analytics.
- Proficient in Time Series Forecasting and predictive modeling using ARIMA, RNNs, and LSTMs in Python and R.
- Strong leadership and project management experience in data preparation, web scraping, predictive modeling, and AI-driven business intelligence solutions.
- Practical experience in Deep Learning applications, including image recognition (CNN), stock price prediction (RNN), recommendation systems (autoencoders), and image captioning (CNN-RNN architectures).
- Highly skilled in translating complex business challenges into AI-powered, data-driven solutions, with a track record of driving measurable improvements in performance, automation, and decision-making.
- Expert in multi-cloud MLOps and AI orchestration, integrating AWS SageMaker, Google Vertex AI, and Azure ML Studio for seamless model training, deployment, and monitoring across distributed cloud environments.
- Implemented end-to-end GenAI and RAG pipelines using LangChain, FAISS, Pinecone, and Vector Search APIs, enhancing enterprise knowledge retrieval, semantic search accuracy, and contextual reasoning by over 35%.
- Demonstrated excellence in automating CI/CD pipelines for AI/ML workflows using Docker, Kubernetes, Jenkins, and Terraform, ensuring reproducible, scalable, and cost-efficient ML model delivery in production.

Technical Skills:

Category	Details
Programming Languages	Python, SQL, R, Java, JavaScript, TypeScript, C/C++, HTML, CSS, ReactJS, Next.js, Angular, jQuery
Statistical & Analytical Methods	Statistical Inference, Hypothesis Testing, Confidence Intervals, ANOVA, Chi-Square Tests, Probability Distributions, Correlation & Covariance, Sampling Techniques, Regression Analysis, Time Series Modeling, Experimental Design, Multivariate Analysis
Machine Learning & AI	Supervised & Unsupervised Learning, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Ensemble Methods (XGBoost, LightGBM, Random Forest), SVM, KNN, Naïve Bayes, Regression, Classification, Clustering, Dimensionality Reduction (PCA, t-SNE), Model Interpretability (SHAP, LIME), Recommendation Systems
Deep Learning	TensorFlow, PyTorch, Keras, ANN, CNN, RNN, LSTM, Autoencoders, Transfer Learning (VGG, ResNet, InceptionNet, MobileNet), Object Detection (YOLO, Faster R-CNN), Image Segmentation (U-Net, Mask R-CNN), GANs, VAEs, Diffusion Models, Neural Style Transfer
Natural Language Processing (NLP)	Text Preprocessing, Tokenization, Lemmatization, BoW, TF-IDF, Word Embeddings (Word2Vec, GloVe, FastText), Transformer Models (BERT, GPT, T5), Seq2Seq, Attention Mechanisms, GRU, Bi-LSTM, NER, Sentiment Analysis, Text Summarization, Question Answering, Topic Modeling (LDA), Libraries: SpaCy, NLTK, Hugging Face
Generative AI & LLMops	LLMs (GPT, LLaMA, Mistral, Gemma), LangChain, LangSmith, RAG, LoRA/QLoRA, Graph & Vector Databases (Neo4j, Pinecone, FAISS, Chroma), Ollama, Prompt Engineering & Templates, Document Chunking & Indexing, Fine-tuning & Inference APIs (Groq, Nvidia NIM), Agentic AI Systems, GANs, CycleGAN, StyleGAN
MLOps & Deployment	CI/CD for ML, Model Deployment (REST APIs, Batch), Model Monitoring & Versioning, MLflow, Airflow, Kubeflow, Experiment Tracking (Weights & Biases), Containerization (Docker, Kubernetes), Feature Stores (Feast), Data Versioning (DVC), Cloud ML Platforms (AWS SageMaker, Bedrock, GCP Vertex AI, Azure AI), LLM Evaluation (TruLens, PromptLayer)
Cloud & Big Data	AWS (SageMaker, EC2, Lambda, EMR, S3, Glue, Athena, Bedrock, Redshift, Kinesis, QuickSight, CloudFormation, CloudWatch, IAM) Google Cloud (Vertex AI, BigQuery, Dataflow, Dataproc, Cloud Functions, Pub/Sub) Azure (Databricks, Synapse, Data Factory, Cognitive Services) Big Data & Orchestration: Hadoop, Spark, Hive, HBase, Scala, Kafka, Airflow DevOps & MLOps: Docker, Kubernetes, Terraform, MLflow, Jenkins
Databases & Data Modeling	MySQL, PostgreSQL, MongoDB, Redshift, RDS, Teradata, SQLite, SQLAlchemy, Snowflake, Star & Snowflake Schemas, ER Diagram Design
Data Visualization & BI Tools	Tableau, Power BI, QuickSight, Kibana, Matplotlib, Seaborn, R Shiny, SAS, Stata
ETL & Data Engineering	Informatica PowerCenter, Data Extraction, Cleansing, Transformation, Profiling, Integration, Data Warehousing Concepts
Version Control & Dev Tools	Git, GitHub, GitLab, Docker Compose, Jenkins, VS Code, Jupyter Notebook, Google Colab, Postman, Swagger, Bash, Terraform, Nginx
Operating Systems	Windows, Linux/Unix, macOS
Collaboration & Productivity Tools	Anaconda, MS Office Suite, Excel, Google Sheets, Notion, Trello, Slack, Figma

Education:

- I successfully completed my Bachelor's degree in Computer Science from KKR & KSR Institute of Technology & Sciences 2015.

Certifications:

- Google Cloud Certified: Professional Machine Learning Engineer (Nov 2020 – Nov 2022).
- Microsoft Certified: Azure AI Engineer Associate (Oct 2023 – Oct 2024).

Professional Experience:

Client: Chase Bank , Dallas, TX

Jan 2024 – Till Date

Role: GenAI/AI Engineer

Responsibilities:

- Led full-cycle development of AI/ML applications, overseeing requirement analysis, design, development, migration, and maintenance using the Software Development Lifecycle (SDLC) and modern Python frameworks. Ensured scalable, maintainable solutions aligned with business and data science objectives.
- Developed robust MLOps pipelines on AWS utilizing SageMaker Pipelines, Lambda, Step Functions, and CloudFormation, automating model training, hyperparameter tuning, and CI/CD deployment. Reduced manual model management overhead by over 40% through workflow automation.
- Implemented Retrieval-Augmented Generation (RAG) systems using Google Vertex AI, Google LLM APIs, and LangChain, enabling scalable LLM deployment for intelligent document retrieval, summarization, and enterprise search use cases.
- Architected hybrid search systems integrating vector similarity search (Google Vector Search, FAISS) and keyword-based search (Elasticsearch) to achieve high-precision semantic retrieval. Improved document relevance scoring and reduced search latency by 30%.
- Built real-time model monitoring and alerting pipelines using Amazon CloudWatch, SageMaker Model Monitor, and QuickSight dashboards, tracking drift detection, latency, and key model performance indicators for continuous improvement.
- Developed LLM-based workflows for call transcript summarization and conversation analytics, leveraging GPT-4, LangChain, and vector databases (FAISS/Pinecone) within RAG architectures, improving summarization accuracy and processing time.
- Fine-tuned domain-specific LLMs using Low-Rank Adaptation (LoRA) and PEFT frameworks, with evaluation benchmarks incorporating BLEU, perplexity, accuracy, and human-in-the-loop feedback for precision validation.
- Engineered end-to-end ML pipelines combining AWS SageMaker Pipelines, Step Functions, and Lambda, ensuring reproducibility, scalability, and continuous delivery across model lifecycle stages from experimentation to production.
- Integrated Google Cloud LLMs and Vector Search within NLP pipelines, building intelligent semantic retrieval, Q&A, and hybrid search systems capable of understanding context beyond keyword matching.
- Built advanced fraud detection systems using Support Vector Machines (SVM) and ensemble models in Scikit-learn, leveraging NumPy, SciPy, Matplotlib, and Pandas for feature engineering, data visualization, and performance analysis.
- Utilized AWS services — S3, DynamoDB, Lambda, EC2, SageMaker, Lex, EMR, and Redshift — for large-scale data ingestion, transformation, model training, and deployment, ensuring secure and cost-optimized ML workflows.
- Processed speech-to-text data using AWS Transcribe, followed by tokenization, lemmatization, and stop-word filtering for NLP tasks such as intent recognition and summarization.
- Performed advanced feature engineering including feature scaling, normalization, label encoding, and interaction feature creation using Scikit-learn preprocessing to improve model performance and interpretability.
- Designed analytical data marts and dimensional models using Star and Snowflake schemas, enabling efficient OLAP reporting and trend analysis across large datasets.
- Developed Power BI dashboards and KPIs for tracking model accuracy, drift, and business performance, integrating data from AWS Redshift, Snowflake, and S3 sources for unified visibility.
- Created predictive and classification models using Linear/Logistic Regression, Naïve Bayes, Random Forest, K-Means, KNN, PCA, Ridge/Lasso Regularization, improving business forecasting accuracy and anomaly detection.
- Leveraged AWS SageMaker for training, tuning, deploying, and versioning ML and DL models, implementing experiment tracking, batch inference, and A/B testing workflows.
- Authored comprehensive ETL documentation, including process flowcharts, data flow diagrams, and technical specifications, ensuring cross-team alignment and maintainability of data pipelines.
- Designed OLTP and OLAP data models in 3NF and Star/Snowflake architectures to optimize query performance, reduce redundancy, and support high-speed analytical workloads.
- Performed time-series analysis and data manipulation using Pandas APIs, applying resampling, rolling-window computations, and timestamp-based queries for forecasting and trend detection.
- Applied Deep Learning architectures (RNN, CNN) to operational optimization use cases — including optimum routing for field crews and storm impact severity prediction for energy and utility datasets using XGBoost and neural networks.
- Implemented cost-effective data warehousing solutions using Snowflake SaaS, enabling high scalability, performance optimization, and near-zero maintenance for analytical workloads.
- Developed MDM (Master Data Management) frameworks, defining data mapping, transformation, and cleansing rules across OLTP, ODS, and OLAP systems, ensuring data consistency and referential integrity.
- Conducted A/B testing and algorithmic experiments to refine semantic search algorithms, applying topic modeling (LDA), TF-IDF weighting, and ML-based ranking models for improved relevance and retrieval performance.
- Implemented computer vision solutions for image recognition and anomaly detection using CNN + SVM hybrid architectures, and developed fraud detection systems employing deep CNNs for feature extraction and anomaly scoring.
- Built interactive Power BI dashboards tailored for business stakeholders to track model health, key KPIs, and executive summaries, enhancing visibility and enabling faster decision-making.

Environment: Experienced across the full Software Development Lifecycle (SDLC) with strong hands-on expertise in Python and its core libraries — Scikit-learn, NumPy, SciPy, Matplotlib, and Pandas — for data analysis and model development. Skilled in AWS cloud services such as S3, DynamoDB, Lambda, EC2, SageMaker, Lex, EMR, and Redshift, as well as Snowflake for scalable data warehousing. Proficient in Machine Learning and Deep Learning algorithms including Naive Bayes, Random Forest, K-Means Clustering, KNN, PCA, and Recurrent Neural Networks (RNN). Experienced with data modeling and architecture using OLTP, ODS, OLAP, and 3NF normalization principles. Adept at building insightful Power BI dashboards for business intelligence and decision-making.

Client: Citi Bank , Brandon, Florida.

Aug 2022 to Dec 2023

Role: Gen-AI/AI Engineer

Responsibilities:

- Performed end-to-end data analysis, validation, and cleansing on large, multi-source datasets to ensure data integrity, consistency, and reliability. Utilized Python (Pandas, NumPy, SciPy) for transformation and Scikit-learn for missing data imputation using statistical and predictive methods, resulting in a significant reduction in data noise and bias during model training.
- Developed advanced Generative AI solutions leveraging Hugging Face Transformers and Meta's LLaMA to perform multi-label classification of complex architectural and technical documents. Designed prompt engineering strategies, optimized tokenization workflows, and achieved high-accuracy semantic understanding for structured document labeling.
- Integrated NLP pipelines using LangChain, vector search (FAISS, Pinecone), and semantic embeddings to implement document clustering, similarity scoring, and hybrid search workflows. This enabled context-aware retrieval-augmented generation (RAG) systems that improved information discovery efficiency across large text corpora.
- Applied fine-tuning techniques such as LoRA (Low-Rank Adaptation) and PEFT (Parameter Efficient Fine-Tuning) to enhance LLM performance on domain-specific datasets. Further implemented GraphRAG techniques to generate structured, explainable outputs and improve retrieval precision for enterprise-scale AI applications.
- Designed, trained, and optimized predictive and classification models using advanced algorithms including Logistic Regression, Linear Regression, Lasso, Ridge, Decision Trees, Random Forest, SVM, KNN, XGBoost, K-Means, Naïve Bayes, and Neural Networks. Conducted Social Network Analysis for relational data insights and influence mapping.
- Developed and deployed ML/NLP models using TensorFlow, Keras, and Scikit-learn for classification, sentiment analysis, and document tagging tasks. Implemented cross-validation and hyperparameter tuning for improved generalization and model robustness.
- Leveraged AWS Cloud services — including EC2, S3, Redshift, RDS, Glacier, API Gateway, SNS, SWF, ELB, and EBS — for data storage, orchestration, and model inference pipelines. Built serverless architectures and auto-scaled model endpoints, reducing infrastructure costs and improving deployment efficiency.
- Deployed Flask-based microservices on Docker containers, orchestrated for scalable model serving and integrated through RESTful APIs for consumption by web and analytics platforms.
- Built cloud-based monitoring dashboards using AWS CloudWatch, Google Cloud Monitoring, and Amazon QuickSight to track ML model performance, latency, and drift metrics in real-time, ensuring reliability and continuous optimization.
- Led Agile-based initiatives to improve cross-functional collaboration between Data Science, DevOps, and Product teams. Facilitated sprint planning, backlog grooming, and retrospective meetings to align goals and accelerate project delivery.
- Designed and implemented data warehouse schemas using Star and Snowflake dimensional models, optimizing data retrieval and query performance for OLAP (Online Analytical Processing) and reporting workloads.
- Performed extensive exploratory data analysis (EDA) using Python (Matplotlib, Seaborn) and developed interactive dashboards in Power BI and QuickSight, enabling stakeholders to visualize KPIs and data-driven insights in real time.
- Applied AI/ML techniques in cognitive science and neuroinformatics, developing neurofeedback models to monitor and control brainwave rhythms, contributing to research in neural signal processing and cognitive training systems.
- Executed advanced feature engineering and dimensionality reduction techniques such as PCA, LDA, and Recursive Feature Elimination (RFE) to enhance model performance, reduce overfitting, and speed up computation on large-scale, high-dimensional datasets.
- Implemented ensemble modeling strategies combining Ridge, Lasso, and XGBoost algorithms to predict loan defaults, risk factors, and financial anomalies, achieving enhanced accuracy and interpretability in model predictions.
- Processed and transformed massive datasets using PySpark MLlib, executing distributed data cleaning and transformation pipelines. Integrated deep learning architectures (CNNs, RNNs, and Transformers) for more complex pattern recognition and text understanding tasks.

- Automated Power BI report refresh schedules (hourly and on-demand) to ensure timely insights and reduce manual reporting overhead for business users.
- Applied a combination of supervised and unsupervised learning techniques, including clustering, regression, NLP, and statistical modeling, to address real-world business and engineering challenges, leading to actionable insights and measurable business improvements.

Environment & Tools:

Experienced in the Software Development Lifecycle (SDLC) with strong proficiency in Python and its core libraries (Scikit-learn, NumPy, SciPy, Pandas, Matplotlib). Skilled in Machine Learning and Deep Learning frameworks (TensorFlow, Keras, XGBoost) and algorithms including Naïve Bayes, Random Forest, K-Means, KNN, PCA. Hands-on experience with AWS cloud services (S3, DynamoDB, Lambda, EC2, SageMaker, Lex, EMR, Redshift) and Snowflake for data warehousing. Expertise in data engineering and analytics with OLAP, OLTP, and PySpark for large-scale data processing. Proficient in NLP using NLTK and building Power BI dashboards for business intelligence and reporting.

Client: Cook County Sheriff's Office, Chicago

Oct 2020 to July 2022

Role: Data Scientist

Responsibilities:

- Actively contributed to Agile development processes, facilitating team ceremonies such as stand-ups, sprint planning, backlog grooming, and reviews to drive project efficiency.
- Partnered with Data Engineers and Operations teams to design and implement ETL pipelines, optimizing SQL queries to extract and process data for analytical and ML needs.
- Developed interactive web-based solutions by creating database models, APIs, and views using Python, ensuring seamless integration with backend systems.
- Conducted statistical analysis, including univariate and multivariate techniques, to uncover patterns, correlations, and insights from complex datasets.
- Built and deployed machine learning models (XGBoost, Random Forest, SVM) for applications such as credit scoring, financial risk optimization, and predictive analytics.
- Engineered Snowflake schema-based data models and automated ETL processes to prepare data for ML feature engineering.
- Implemented production-ready ML pipelines using Docker and AWS SageMaker, enabling scalable model deployment and inference.
- Designed hybrid search strategies, integrating traditional SQL queries with semantic search and NLP pipelines to enhance document retrieval and clustering.
- Developed end-to-end data engineering workflows on AWS (S3, RDS) and Snowflake, ensuring efficient and reliable data ingestion.
- Conducted feasibility and cost analyses during requirement gathering, reducing project expenses by up to 60%.
- Applied data cleansing, validation, and imputation techniques with Pandas, NumPy, and Scikit-learn, ensuring high-quality datasets for modeling.
- Leveraged NLP and AI tools to improve text classification and sentiment analysis, minimizing false positives on unstructured and semi-structured data.
- Delivered SEO-optimized solutions and BI dashboards, improving online visibility and business performance by 70%.
- Built time-series models and optimization solutions in collaboration with analytics teams to support forecasting and decision-making.
- Developed a multilingual e-learning management system, implementing SCORM-compliant solutions with Articulate 360 and Redwood Web Authoring Tools.
- Created logical and physical data models using Star and Snowflake schemas with Erwin, supporting data warehousing and business intelligence needs.
- Designed interactive dashboards and reports using Power BI and Tableau, transforming complex datasets into actionable insights for stakeholders.

Key Tools & Technologies: Python, SQL, PL/SQL, Pandas, NumPy, SciPy, Scikit-learn, NLTK, XGBoost, Matplotlib, Seaborn, APIs, ER/Studio, Erwin, Snowflake, Redshift, Teradata, Oracle, AWS (S3, RDS), OLAP, OLTP, Docker, Power BI, Power Query, Tableau, ETL Pipelines.

Client: Sam's Club , Tampa, Florida.

Feb 2018 to Sep 2020

Role: Data Analyst**Responsibilities:**

- Conducted data analysis, migration, and preparation for customer segmentation and profiling, enabling more targeted business insights.
- Implemented statistical calculations and predictive modeling in Python using Pandas, NumPy, Seaborn, SciPy, Matplotlib, Scikit-learn, and NLTK.
- Designed and maintained data warehousing and data modeling procedures, building ETL pipelines to integrate data from multiple sources efficiently.
- Developed scalable Python algorithms for data mining and predictive modeling, leveraging advanced statistical techniques to support decision-making.
- Utilized ETL tools to rapidly deploy pipelines for data gathering, cleansing, and preparation, optimizing workflows for multiple datasets.
- Created multivariate data validation scripts in Python for equity, derivatives, currency, and commodity data, improving pipeline efficiency by 17%.
- Applied predictive analytics to design sampling methodologies and perform pricing analysis for client products.
- Optimized ETL processes transitioning workflows from Alteryx to Snowflake, improving performance and scalability.
- Developed data visualizations and dashboards using Tableau and advanced MS Excel (macros, pivot tables, lookups, arrays) to provide actionable business insights.
- Leveraged Python, Mahout, Hadoop, and MongoDB for data analytics and automation, processing large-scale structured and unstructured datasets.
- Managed GIT repositories, ensuring version control, access management, and support for collaborative development.
- Fostered team collaboration and effectively managed competing priorities across weekly, bi-weekly, monthly, and quarterly deliverables.
- Worked extensively with ER/Studio on OLAP and OLTP applications to design and manage enterprise data models.
- Generated ad-hoc SQL queries (joins, transformations) to extract and manipulate data from Oracle and SQL Server databases.
- Analyzed business requirements, enhanced functional specifications, conducted testing, and resolved critical bugs in a Learning Management System (LMS).
- Built and deployed a web-based E-Learning application using jQuery, JavaScript, HTML, and Node.js, enabling interactive learning experiences.
- Performed data cleaning and transformation in Python and developed KPI dashboards and reports in Tableau.
- Published interactive dashboards and reports from Tableau Desktop to Tableau Server for enterprise-wide access and reporting.

Environment & Tools: Python, Pandas, NumPy, Seaborn, SciPy, Matplotlib, Scikit-learn, NLTK, ETL, Alteryx, Snowflake, Tableau, jQuery, JavaScript, HTML, Node.js, Hadoop, MongoDB, OLTP, OLAP, ER Studio, Oracle, SQL Server, SQL, Tableau Server.

Client: Tata Consultancy Services Pvt Ltd, Hyderabad, India

July 2015 to Aug 2017

Role: Data Engineer**Responsibilities:**

- Translated complex business and analytical requirements into detailed technical specifications, collaborating with cross-functional teams and business users to design and develop interactive Tableau dashboards and automated reporting systems. This helped leadership teams make data-driven decisions quickly and confidently.
- Built end-to-end automated reporting workflows in Tableau, significantly reducing manual report generation time by over 40%. Designed dashboards that supported dynamic parameterization, real-time filtering, and drill-down capabilities for improved decision-making and self-service analytics.
- Partnered with data engineering, product management, and healthcare domain experts to understand business objectives and integrate clinical, pharmacy, claims, and membership data into unified analytical views. Delivered insights that guided strategic initiatives in healthcare product design, member engagement, and cost optimization.
- Developed advanced SAS scripts and analytical models to perform statistical studies, validate hypotheses, and conduct recurring data analysis for both internal business intelligence teams and external clients. Automated several repetitive SAS tasks, improving analytical efficiency and accuracy.
- Led data quality assurance initiatives by performing comprehensive data validation, hygiene checks, and consistency verification across multiple data pipelines. Designed validation programs to flag anomalies, correct inconsistencies, and ensure high data reliability across all reporting layers.

- Analyzed complex healthcare datasets including medical claims, pharmacy transactions, wellness programs, and membership details to uncover trends, identify high-risk populations, and support value-based care initiatives.
- Consolidated and integrated data from various functional systems — such as membership, claims/encounter, pharmacy, lab, and radiology data sources — to create a unified analytical dataset, enabling comprehensive ad-hoc reporting and proposal generation for large enterprise clients.
- Managed and optimized Tableau data connections, ensuring accurate and timely refreshes from sources like SQL Server, Oracle, and Redshift. Maintained a structured approach to data versioning, incremental loads, and access permissions for smooth operational workflows.
- Specialized in metadata management, defining dimensions, hierarchies, measures, and calculated fields within Tableau to maintain standardization, reduce redundancy, and ensure consistent reporting across business units.
- Designed advanced Tableau visualizations including bar, line, pie, tree map, bubble, waterfall, bump, funnel, and Bollinger charts, each tailored to specific KPIs and stakeholder requirements. Implemented interactive filters, parameters, and tooltips to enhance user engagement and usability.
- Applied SQL and data blending techniques to merge and transform disparate data sources within Tableau, enabling hybrid data models that combine structured SQL data with semi-structured datasets from other systems.
- Implemented OLTP and OLAP data models to support both real-time transactions and analytical workloads. Created efficient Star and Snowflake schemas for structured data representation and faster analytical query performance.
- Provided expert consultation on healthcare data analytics and visualization best practices, helping teams interpret insights from claims and pharmacy data to optimize operational and clinical outcomes.
- Collaborated in Agile/Scrum environments, participating in daily stand-ups, sprint planning, and review meetings, ensuring on-time delivery of dashboard enhancements and reporting features.
- Built a feedback-driven dashboard optimization framework, where stakeholders could log enhancement requests and view live updates—reducing communication gaps and improving product adoption rates.
- Documented workflows, data sources, and business logic, ensuring full transparency and easy maintainability for future Tableau developers and analysts.

Environment: SQL, Python, Tableau, Hive, SAS, Pig, MS Office, Ms Excel, Windows.